ANFENG XU

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EDUCATION

University of Southern California (USC) Expected Dec 2026
Ph.D. in Electrical Engineering, Minor in Computer Science GPA: 3.96/4.0

University of Southern California (USC) May 2024
M.S. in Electrical Engineering GPA: 4.0/4.0

University of California, San Diego (UCSD) June 2021
B.S. in Electrical Engineering, Minor in Mathematics GPA: 3.97/4.0

RESEARCH INTERESTS

Speech Processing, Multimodal Learning, Generative AI, Healthcare AI, LLM.

WORK EXPERIENCE

Meta Superintelligence Labs May 2025 - Aug 2025
Ph.D. Research Scientist Intern, Advisors: Yashesh Gaur, Naoyuki Kanda *Seattle*

Developed speech naturalness prediction models for conversational speech generative models (e.g., LLaMA Full-Duplex).
Built a large-scale synthetic data augmentation pipeline using internal speech generative models.

Meta Reality Labs May 2024 - Aug 2024
Ph.D. Research Scientist Intern, Advisor: Biqiao Zhang *Seattle*

Conducted research to enhance the Keyword Spotting (KWS) models for wearable devices.

Signal Analysis and Interpretation Laboratory (SAIL), USC August 2021 - Present
Ph.D. Research Assistant, Advisor: Shrikanth Narayanan *Los Angeles*

Conduct research projects with the main focus on speech processing and multi-modality.
Lead the child speech/video project, mentor MS students, and manage the IEMOCAP/CreativeIT data distribution.

RESEARCH PROJECTS

Joint ASR and Speaker Diarization for Child-Adult Interactions (USC - BU - Apple) Oct 2024 - Present

Developed LLM error correction methods for ASR for child-adult speech.
Developed an efficient ASR model with integrated speaker and timestamp prediction using serialized output training, jointly trained for speaker diarization.

Speaker Diarization for Child-Adult Interactions (USC - BU - Apple) Jan 2022 - Oct 2024

Proposed annotation framework and modeling pipeline to predict language capabilities of children directly from audio.
Developed an audio-visual child-adult speaker classification method.
Built a child-adult speaker diarization model with Speech Foundation Models.
Built simulated conversations from AudioSet to further enhance child-adult speaker diarization.

AWARDS

ECE Ph.D. Screening Exam (USC, 2022): Ranked 1st for the Signals and Systems track.

Annenberg Fellowship (USC, 2021): Top-off fellowship selected among incoming Ph.D. students.

Summa Cum Laude (UCSD, 2021): GPA within the top 2% of the graduating class.

LANGUAGES

English (fluent), Japanese (native), Mandarin Chinese (conversational)

SELECTED PUBLICATIONS

[KDD'26] T Feng, K Huang, **A Xu**, et al. "Voxlect: A Speech Foundation Model Benchmark for Modeling Dialects and Regional Languages Around the Globe."

[Interspeech'25] T Feng, **A Xu**, X Shi, S Bishop, S Narayanan. "Egocentric speaker classification in child-adult dyadic interactions: From sensing to computational modeling."

[Interspeech'25] Z Shi, X Shi, **A Xu**, T Feng, H Srivastava, S Narayanan, M Mataric. "Examining Test-Time Adaptation for Personalized Child Speech Recognition."

[Interspeech'25] **A Xu***, T Feng*, SH Kim, S Bishop, C Lord, S Narayanan. "Large Language Models based ASR Error Correction for Child Conversations."

[ICASSP'25] **A Xu**, T Feng, H Tager-Flusberg, C Lord, S Narayanan. "Data Efficient Child-Adult Speaker Diarization with Simulated Conversations."

[ICASSP'25] **A Xu**, B Zhang, S Kong, Y Huang, Z Yang, S Srivastava, M Sun. "Effective Integration of KAN for Keyword Spotting."

[NeurIPS Workshop'24] (GenAI for Health) T Feng, **A Xu**, et al. "Can Generic LLMs Help Analyze Child-Adult Interactions Involving Children with Autism in Clinical Observation?"

[Interspeech'24] **A Xu**, K Huang, T Feng, L Shen, H Tager-Feng, S Narayanan. "Exploring Speech Foundation Models for Speaker Diarization in Child-Adult Dyadic Interactions."

[ICASSP'24] **A Xu**, K Huang, T Feng, H Tager-Feng, S Narayanan. "Audio-visual child-adult speaker classification in dyadic interactions."

[ACM MM'23] D Bose, R Hebber, T Feng, K Somandepalli, **A Xu**, S Narayanan. "MM-AU:Towards Multimodal understanding of advertisement videos."

[Interspeech'23] **A Xu**, R Hebbar*, R Lahiri*, T Feng*, L Butler, L Shen, H Tager-Flusberg, and S Narayanan. "Understanding Spoken Language Development of Children with ASD Using Pre-trained Speech Embeddings."

(* equal contribution)

MANUSCRIPTS UNDER REVIEW

A Xu, T Feng, S Bishop, C Lord, S Narayanan. "End-to-End Joint ASR and Speaker Role Diarization with Child-Adult Interactions," under review for TASLP.

A Xu, Y Gaur, N Kanda, Z Ouyang, K Žmolíková, D Raj, S Merello, A Sun, O Kalinli. "Conversational Speech Naturalness Predictor," under internal review for Interspeech 2026.

T Feng, **A Xu**, J Lee, S Narayanan. "VoxCog: Towards End-to-End Multilingual Cognitive Impairment Classification through Dialectal Knowledge," under review for Computer Speech & Language.

T Feng, J Lee, **A Xu**, et al. "Vox-Profile: A Speech Foundation Model Benchmark for Characterizing Diverse Speaker Traits and Speech Properties," under review for DMLR.